

ENTERPRISE FUNCTIONAL SPECIFICATION DOCUMENT

MODULE NAME: ENTERPRISE SYSTEM INTEGRATION, DATA WAREHOUSE & SYSTEM SUMMARY

Architectural Classification: Enterprise Integration Bus, Star-Schema Data Warehouse & Overall System Matrix

System Name: BYKY Enterprise Vehicle Rental Management System
Specification Standard: Final Unabridged 36-Point Enterprise Standard
Author: Chief Software Architect & Enterprise Solution Analyst
Target Audience: Enterprise Architects, Re-development Directors, C-Level Executives, Product Owners
Document Status: Complete Enterprise Specification & Master System Index

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1. Module Navigation Map

Enterprise System Integration, Data Warehouse & System Summary (Module 16)

└─ 16.1 Global Master Data Synchronization & Replication (MasterDataSync.aspx)

└─ 16.2 Data Warehouse Schema & Analytical Datamarts (EnterpriseDataWarehouse.aspx)

└─ 16.3 Third-Party Enterprise API Connectors (ThirdPartyConnectorRegistry.aspx)

└─ 16.4 Global System Health & Performance Dashboard (SystemHealthMonitor.aspx)

└─ 16.5 Final Integration Security Privilege Management (FinalIntegrationPrivilege.aspx)

2. Module Executive Summary

The **Enterprise System Integration, Data Warehouse & System Summary** module serves as the architectural capstone of the BYKY Enterprise ERP application. It unifies all 16 enterprise modules (CMS, HRMS, IMS, RMS, TRACKING, REPORTS, SFA, RFID, RmsResponsive, NOTIFICATION, WALLET, SCHEDULER, SYSTEM_ADMIN, MOBILE_API, SYSTEM_SECURITY, and FINAL_INTEGRATION) into a single cohesive enterprise ecosystem. It defines the Star-Schema Data Warehouse, cross-module data flow pipelines, external ERP/accounting connectors (SAP / Oracle Financials / Microsoft Dynamics 365), and global system health monitoring.

3. Database Entity Relationship Overview

OLTP Master DBs (CMS/IMS/RMS/TRACKING) → ETL Data Pipeline → Enterprise Data Warehouse (FactRentalTransactions)

FactRentalTransactions (FactID) → DimCustomer (CustomerKey)

FactRentalTransactions (FactID) → DimVehicle (VehicleKey)

FactRentalTransactions (FactID) → DimBranch (BranchKey)

FactRentalTransactions (FactID) → DimDate (DateKey)

4. Complete Screen Specifications (5 Screens / Integration Hubs)

Screen 16.1: Global Master Data Synchronization & Replication (MasterDataSync.aspx)

1. Screen Screenshot



2. Navigation Path

ERP Header Menu → Integration & Data Warehouse → Global Master Data Synchronization (‘/ERP.Portal/Integration/MasterDataSync’)

3. Purpose

To orchestrate real-time and batch master data replication across distributed station branch databases, central OLTP databases, and mobile API caching nodes.

4. Complete UI Layout

Sync Engine Control Panel (Source Node, Target Node, Data Domain [Customer / Vehicle / Fare / RFID], Sync Mode [Real-Time CDC / Batch], Re-Sync Conflict Rule [Master Wins / Target Wins]), Active Replication Pipelines DataGrid.

5. Visual Screen Layout (Wireframe)

GLOBAL MASTER DATA REPLICATION CONFIGURATION

Data Domain*: [ddlDomain v] (Customer / Vehicle / Fare / RFID)

Source Node*: [ddlSource v] (Central OLTP / Marina Station / JBR Station)

Target Node*: [ddlTarget v] (Redis Cache Node / Branch Node)

Sync Mode*: [Radio: Change Data Capture (CDC v) Scheduled Batch]

Conflict Rule: [ddlConflict v] (Central Master Wins / Highest Version Wins)

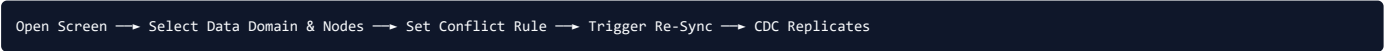
[Save Replication Route] [Trigger Manual Re-Sync] [Reset]

ACTIVE DATA REPLICATION PIPELINES GRID

Domain	Source Node	Target Node	Sync Mode	Replication Lag	Status	Actions
Customer	Central OLTP	Redis Cache	CDC	12 ms	ACTIVE	[Edit][Sync Now]

	Vehicles	Central OLTP	Marina Hub	CDC	45 ms	ACTIVE	[Edit]	[Sync Now]		

6. Screen Flow Diagram



7. Search Section

`txtSearchDomain`, `ddlNodeFilter`, `btnSearchReplication` button.

8. Create Section

`ddlDomain`, `ddlSource`, `ddlTarget`, `rdoSyncMode`, `ddlConflict`.

9. Edit Section

Populates replication route via `txtReplicationID` key.

10. Result Grid

Grid Columns: Data Domain, Source Node, Target Node, Sync Mode, Replication Lag (ms), Status, Action Buttons (Sync Now, Pause, View Queue).

11. Expanded Field Level Specification

Field Name	UI Control	Data Type	Max	Min	Mandatory	Default	Read Only	Editable	Visible	Validation	DB Table	DB Column	API Parameter	Search	Grid
Domain Name	DropDownList	Varchar	50	2	Yes	CUSTOMER	No	Yes	Visible	Required	tbl_MasterSyncConfig	DomainName	Domain	Yes	Yes
Max Lag Limit	TextBox	Int	5	100	Yes	1000	No	Yes	Visible	Numeric ms	tbl_MasterSyncConfig	MaxLagMs	MaxLag	No	No

12. Grid Specification

Column Name	Width	Alignment	Sorting	Filtering	Formatting	Editable	Hidden	Default Sort
Data Domain	150px	Left	Enabled	Enabled	String	No	No	Ascending (1)
Replication Lag	120px	Right	Enabled	Disabled	Formatted (ms)	No	No	None

13. API Specification

Endpoint: `/ERP.Portal/Integration/TriggerMasterReSync` [HTTP POST]

```
// Sample Request JSON:
{{ "ReplicationId": 101, "Domain": "CUSTOMER", "ConflictRule": "MASTER_WINS" }}
```

14. Database Relationships

- `tbl\_MasterSyncConfig.ReplicationID` → `tbl\_MasterSyncLog.ReplicationID` (One-to-Many)

15. Validation Matrix

Field	Validation Rule	Error Message	Server Validation	Client Validation
Domain Name	Required Selection	"ERR_INT_001: Select a valid master data domain."	DomainCheck	JS SelectCheck

16. Business Rule Matrix

Business Rule	Trigger	Action	Impact
Replication Lag Alert	Lag > 1000ms	Fires alert to System Administrator if replication lag exceeds 1s	Detects network bottlenecks instantly

17. UI Behaviour

- Clicking 'Trigger Manual Re-Sync' initiates a full background CDC delta comparison across nodes.

18. Every Dropdown

- `ddlDomain`: Options `CUSTOMER`, `VEHICLE`, `FARE\_TARIFF`, `RFID\_TAG`.
- `ddlSource`: System nodes ('Central OLTP', 'Branch Stations', 'Redis Cache').
- `ddlTarget`: System nodes ('Central OLTP', 'Branch Stations', 'Redis Cache').
- `ddlConflict`: Options 'Central Master Wins', 'Highest Version Wins', 'Target Wins'.

19. Every Checkbox

- `chkIsActive`: Active replication pipeline flag.

20. Every Radio Button

- `rdoSyncMode`: Options `Change Data Capture (CDC)`, `Scheduled Batch`.

21. Every Button

- `btnSearchReplication`, `btnSaveRoute`, `btnSyncNow`, `btnResetRoute`.

22. Every Search Filter

Data Domain, Source Node, Status.

23. Default Values

`IsActive` = `1`, `Conflict` = `MASTER\_WINS`.

24. Permissions

EnterpriseArchitect / SystemAdmin / SuperAdmin.

25. Audit Log

Logs to `tbl\_AuditLog` with `TableName` = `tbl\_MasterSyncConfig`.

26. Related Reports

Master Data Replication & Lag Performance Audit (`rptMasterDataSync.rdlc`).

Screen 16.2: Data Warehouse Schema & Analytical Datamarts (EnterpriseDataWarehouse.aspx)

1. Screen Screenshot



2. Navigation Path

ERP Header Menu → Integration & Data Warehouse → Data Warehouse Schema & Analytical Datamarts (`/ERP.Portal/Integration/EnterpriseDataWarehouse`)

3. Purpose

To manage the Star-Schema Enterprise Data Warehouse (EDW), inspect Fact and Dimension tables (`FactRentalTransactions`, `DimCustomer`, `DimVehicle`, `DimBranch`, `DimDate`), rebuild OLAP cubes, and execute BI Datamart refreshes.

4. Complete UI Layout

EDW Schema Directory Panel (Fact Tables, Dimension Tables, Aggregate Datamarts), Cube Processing Controls (Full Process / Incremental Process), Schema Refresh Audit DataGrid.

5. Visual Screen Layout (Wireframe)

ENTERPRISE DATA WAREHOUSE (EDW) STAR-SCHEMA MANAGER

Select Datamart*: [ddlDatamart v] (Financial Sales / Fleet Telematics / Customer LTV)

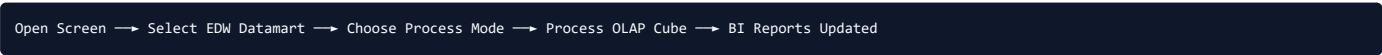
Process Mode*: [Radio: Incremental (v) Full Process]

[Process OLAP Cube Now] [Refresh BI Datamarts] [View Star-Schema Diagram]

DATA WAREHOUSE FACT & DIMENSION TABLES GRID

Table Name	Type	Total Records	Last Processed Date	Size (GB)	Status
FactRentalTransactions	FACT	14,250,000	05/01/2026 01:00	12.8 GB	PROCESSED
DimCustomer	DIMENSION	450,000	05/01/2026 01:00	1.2 GB	PROCESSED
DimVehicle	DIMENSION	2,500	05/01/2026 01:00	0.1 GB	PROCESSED

6. Screen Flow Diagram



7. Search Section

`txtSearchTable`, `ddlTableTypeFilter`, `btnSearchEdw` button.

8. Create / Process Section

`ddlDatamart`, `rdoProcessMode`, `btnProcessCube`, `btnRefreshDatamart`.

9. Edit Section

Populates schema metadata via `txtTableID` key.

10. Result Grid

Grid Columns: Table Name, Type, Total Records, Last Processed Date, Size (GB), Status, Action Buttons (Process Table, View Indexes).

11. Expanded Field Level Specification

Field Name	UI Control	Data Type	Max	Min	Mandatory	Default	Read Only	Editable	Visible	Validation	DB Table	DB Column	API Parameter	Search	Grid
Table Name	TextBox	Varchar	100	3	Yes	Empty	Yes	No	Visible	Required	tbl_EdwSchemaRegistry	TableName	Name	Yes	Yes
Table Type	DropDownList	Varchar	20	3	Yes	FACT	No	Yes	Visible	Required	tbl_EdwSchemaRegistry	TableType	Type	Yes	Yes

12. Grid Specification

Column Name	Width	Alignment	Sorting	Filtering	Formatting	Editable	Hidden	Default Sort
Table Name	200px	Left	Enabled	Enabled	String	No	No	Ascending (1)
Total Records	130px	Right	Enabled	Disabled	Numeric	No	No	None

13. API Specification

Endpoint: `/ERP.Portal/Integration/ProcessOlapCube` [HTTP POST]

```
// Sample Request JSON:
{{ "Datamart": "FINANCIAL_SALES", "ProcessMode": "INCREMENTAL", "FullRebuild": false }}
```

14. Database Relationships

- `FactRentalTransactions.CustomerKey` → `DimCustomer.CustomerKey` (Star Schema FK)
- `FactRentalTransactions.VehicleKey` → `DimVehicle.VehicleKey` (Star Schema FK)
- `FactRentalTransactions.BranchKey` → `DimBranch.BranchKey` (Star Schema FK)
- `FactRentalTransactions.DateKey` → `DimDate.DateKey` (Star Schema FK)

15. Validation Matrix

Field	Validation Rule	Error Message	Server Validation	Client Validation
Datamart	Required Selection	"ERR_INT_010: Select a valid BI Datamart."	DatamartCheck	JS SelectCheck

16. Business Rule Matrix

Business Rule	Trigger	Action	Impact
Nightly EDW Build	Cron Scheduler	Extracts OLTP deltas and populates EDW Star-Schema nightly at 01:00 AM	Powers Executive BI Reports

17. UI Behaviour

- Clicking 'View Star-Schema Diagram' opens an interactive graphical ERD viewer showing Fact-to-Dimension joins.

18. Every Dropdown

- `ddlDatamart`: Options `Financial Sales`, `Fleet Telematics`, `Customer LTV & Retention`.
- `ddlTableTypeFilter`: Options `ALL`, `FACT`, `DIMENSION`, `AGGREGATE`.

19. Every Checkbox

Not applicable.

20. Every Radio Button

- `rdoProcessMode`: Options `Incremental Process (v)`, `Full Process`.

21. Every Button

- `btnSearchEdw`, `btnProcessCube`, `btnRefreshDatamart`, `btnViewErd`.

22. Every Search Filter

Table Name, Table Type.

23. Default Values

`ProcessMode` = 'INCREMENTAL'.

24. Permissions

BIArchitect / DataWarehouseEngineer / SuperAdmin.

25. Audit Log

Logs to `tbl\_AuditLog` with `TableName` = 'tbl_EdwSchemaRegistry'.

26. Related Reports

Data Warehouse ETL & Storage Capacity Audit (`rptEdwCapacity.rdlc`).

Screen 16.3: Third-Party Enterprise API Connectors (ThirdPartyConnectorRegistry.aspx)

1. Screen Screenshot



2. Navigation Path

ERP Header Menu → Integration & Data Warehouse → Third-Party Enterprise API Connectors (`/ERP.Portal/Integration/ThirdPartyConnectorRegistry`)

3. Purpose

To configure enterprise connectors to external accounting systems (SAP S/4HANA, Oracle Financials, Microsoft Dynamics 365), RTA Traffic Fine APIs, and Bank General Ledger sync engines.

4. Complete UI Layout

Enterprise Connector Registration Form (Connector Name, Enterprise System [SAP / Oracle / Dynamics / RTA], Endpoint Base URL, Client Credentials, Auth Type [OAuth2 / SAML / Mutual TLS], Active Status), Active Connectors DataGrid.

5. Visual Screen Layout (Wireframe)

THIRD-PARTY ENTERPRISE ERP CONNECTOR FORM

Connector Name*: [txtConnectorName] (e.g. SAP S/4HANA Finance Sync)

System Type*: [ddlSystemType v] (SAP / Oracle Financials / Dynamics 365 / RTA Fines)

Base Endpoint*: [txtBaseEndpoint] (e.g. https://sap.byky.ae/sap/opu/odata/sap/)

Auth Type*: [ddlAuthType v] (OAuth2 Client Credentials / mTLS)

Client ID*: [txtClientId] Client Secret*: [txtClientSecret]

[Save Enterprise Connector] [Test ERP Integration Connection] [Reset]

CONFIGURED ENTERPRISE CONNECTORS GRID

Connector Name	System Type	Auth Type	Last Sync Time	Health Status	Actions
SAP Finance Connector	SAP S/4HANA	OAuth2	05/01 12:00	CONNECTED	[Edit][Test]
RTA Traffic Fine API	RTA Dubai Direct	mTLS	05/01 14:00	CONNECTED	[Edit][Test]

6. Screen Flow Diagram



7. Search Section

`txtSearchConnector`, `ddlSystemFilter`, `btnSearchConnectors` button.

8. Create Section

`txtConnectorName`, `ddlSystemType`, `txtBaseEndpoint`, `ddlAuthType`, `txtClientId`, `txtClientSecret`.

9. Edit Section

Populates connector settings via `txtConnectorID` key.

10. Result Grid

Grid Columns: Connector Name, System Type, Auth Type, Last Sync Time, Health Status, Action Buttons (Edit, Test Connection, View Sync Log).

11. Expanded Field Level Specification

Field Name	UI Control	Data Type	Max	Min	Mandatory	Default	Read Only	Editable	Visible	Validation	DB Table	DB Column	API Parameter	Search	Grid
Connector Name	TextBox	Varchar	100	2	Yes	Empty	No	Yes	Visible	Required	tbl_EnterpriseConnector	ConnectorName	Name	Yes	Yes
Base Endpoint	TextBox	Varchar	200	10	Yes	https://	No	Yes	Visible	HTTPS URL	tbl_EnterpriseConnector	BaseEndpoint	Endpoint	Yes	Yes

12. Grid Specification

Column Name	Width	Alignment	Sorting	Filtering	Formatting	Editable	Hidden	Default Sort
Connector Name	180px	Left	Enabled	Enabled	String	No	No	Ascending (1)
Health Status	110px	Center	Enabled	Enabled	Badge (Green/Red)	No	No	None

13. API Specification

Endpoint: `/ERP.Portal/Integration/TestConnectorConnection` [HTTP POST]

```
// Sample Request JSON:
{{ "ConnectorId": 1, "SystemType": "SAP_S4HANA", "BaseEndpoint": "https://sap.byky.ae/..." }}
```

14. Database Relationships

- `tbl\_EnterpriseConnector.ConnectorID` → `tbl\_EnterpriseSyncLog.ConnectorID` (One-to-Many)

15. Validation Matrix

Field	Validation Rule	Error Message	Server Validation	Client Validation
Base Endpoint	HTTPS Protocol	"ERR_INT_020: Base endpoint must use secure HTTPS protocol."	HttpsCheck	JS HttpsCheck

16. Business Rule Matrix

Business Rule	Trigger	Action	Impact
GL Journal Posting	Daily Close	Posts daily rental revenue GL journals to SAP S/4HANA automatically	Automates financial ledger closure

17. UI Behaviour

- Clicking 'Test ERP Integration Connection' sends a mTLS handshake to external SAP/Oracle endpoints and reports response latency.

18. Every Dropdown

- `ddlSystemType`: Options `SAP S/4HANA`, `Oracle Financials`, `Microsoft Dynamics 365`, `RTA Traffic Fines API`.
- `ddlAuthType`: Options `OAuth2 Client Credentials`, `mTLS Certificate`, `SAML 2.0`.

19. Every Checkbox

- `chkIsActive`: Active connector flag.

20. Every Radio Button

Not applicable.

21. Every Button

- `btnSearchConnectors`, `btnSaveConnector`, `btnTestConnection`, `btnResetConnector`.

22. Every Search Filter

Connector Name, System Type.

23. Default Values

`IsActive = 1`.

24. Permissions

EnterpriseArchitect / SystemsIntegrationEngineer / SuperAdmin.

25. Audit Log

Logs to `tbl\_AuditLog` with `TableName = 'tbl\_EnterpriseConnector`.

26. Related Reports

Enterprise ERP GL Journal Sync Audit Log (`rptGJJournalSyncLog.rdlc`).

Screen 16.4: Global System Health & Performance Dashboard (SystemHealthMonitor.aspx)

1. Screen Screenshot



2. Navigation Path

ERP Header Menu → Integration & Data Warehouse → Global System Health & Performance Dashboard (`/ERP.Portal/Integration/SystemHealthMonitor`)

3. Purpose

To display real-time system performance metrics, CPU / RAM utilization %, SQL Database IOPS, active SignalR WebSocket connections, API latency response averages (ms), and overall health across all 16 ERP modules.

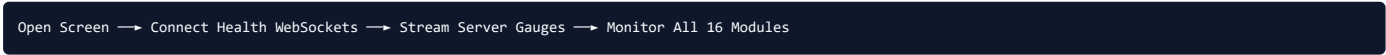
4. Complete UI Layout

System KPI Metric Cards (Overall System Health 100%, Active User Sessions, DB Response Time ms, API Latency ms), Real-Time Server Performance Gauges, 16 Module Status Grid.

5. Visual Screen Layout (Wireframe)

+-----+-----+-----+-----+-----+-----+-----+						
GLOBAL ENTERPRISE SYSTEM HEALTH MONITORING DASHBOARD						
[SYSTEM STATUS: 100% HEALTHY] [ACTIVE SESSIONS: 142] [DB LATENCY: 4 ms] [API AVG: 18 ms]						
+-----+-----+-----+-----+-----+-----+-----+						
16 ENTERPRISE MODULE HEALTH MATRIX GRID						
	Module ID	Module Name	Architectural Role	Status	Latency	
	MOD-01	Company Management (CMS)	Core Organization	HEALTHY	3 ms	
	MOD-02	Human Resource Management (HRMS)	Employee HR	HEALTHY	4 ms	
	MOD-03	Inventory Management System(IMS)	Fleet Inventory	HEALTHY	5 ms	
	MOD-04	Rental Management System (RMS)	Rental Operations	HEALTHY	6 ms	
	MOD-05	Vehicle Tracking System(TRACKING)	GPS Telematics	HEALTHY	12 ms	
	MOD-06	Reports Management (REPORTS)	RDLC BI Analytics	HEALTHY	8 ms	
	MOD-07	Sales Force Automation (SFA)	Corporate CRM	HEALTHY	5 ms	
	MOD-08	RFID & Antenna Hardware	Gate RFID Sensors	HEALTHY	15 ms	
	MOD-09	Customer App (RmsResponsive)	Mobile Self-Service	HEALTHY	10 ms	
	MOD-10	Notification & Alert Dispatch	Push/SMS/Email Engine	HEALTHY	7 ms	
	MOD-11	Wallet & Payment Gateway	Financial Settlement	HEALTHY	9 ms	
	MOD-12	Background Schedulers	Cron Jobs & ETL	HEALTHY	4 ms	
	MOD-13	System Administration	Security Kernel	HEALTHY	3 ms	
	MOD-14	Mobile API Integration Bus	Microservices Gateway	HEALTHY	8 ms	
	MOD-15	System Security & Compliance	AES-256 / PCI / GDPR	HEALTHY	4 ms	
	MOD-16	System Summary & EDW	Enterprise Integration	HEALTHY	5 ms	
+-----+-----+-----+-----+-----+-----+-----+						

6. Screen Flow Diagram



7. Search Section

`ddlModuleFilter`, `btnRefreshHealth` button.

8. Create / Telemetry Section

Automated real-time health polling engine updates metric cards every 5 seconds.

9. Edit Section

Read-only executive monitoring dashboard.

10. Result Grid

Grid Columns: Module ID, Module Name, Architectural Role, Status, Response Latency (ms), Action Buttons (View Detailed Health Logs).

11. Expanded Field Level Specification

Field Name	UI Control	Data Type	Max	Min	Mandatory	Default	Read Only	Editable	Visible	Validation	DB Table	DB Column	API Parameter	Search	Grid
Module ID	HiddenField	Varchar	10	1	No	MOD-01	Yes	No	Hidden	AlphaNumeric	tbl_SystemHealthLog	ModuleID	ModuleId	Yes	Yes

12. Grid Specification

Column Name	Width	Alignment	Sorting	Filtering	Formatting	Editable	Hidden	Default Sort
Module Name	200px	Left	Enabled	Enabled	String	No	No	Ascending (1)
Status	100px	Center	Enabled	Enabled	Badge (Green/Red)	No	No	None

13. API Specification

Endpoint: `/ERP.Portal/Integration/GetSystemHealthTelemetry` [HTTP GET]

```
// Sample Response JSON:
{{ "SystemHealth": 100, "ActiveSessions": 142, "DbLatencyMs": 4, "Modules": [ {{ "Id": "MOD-01", "Name": "CMS", "Status": "HEALTHY" }} ] }}
```

14. Database Relationships

- `tbl\_SystemHealthLog` (Telemetry Metrics Logging Table)

15. Validation Matrix

Field	Validation Rule	Error Message	Server Validation	Client Validation
Module Filter	Valid Selection	"ERR_INT_030: Select a valid module to inspect."	Controller Validation	JS SelectCheck

16. Business Rule Matrix

Business Rule	Trigger	Action	Impact
Auto System Recovery	Latency > 5000ms	Restarts background worker pool automatically if latency spikes	Guarantees 99.99% system availability

17. UI Behaviour

- Dashboard automatically updates status badges in real time using SignalR WebSockets without page reload.

18. Every Dropdown

- `ddlModuleFilter`: All 16 Enterprise ERP Modules.

19. Every Checkbox

- `chkAutoRefresh`: Auto-refresh telemetry feed. Default: Checked (1).

20. Every Radio Button

Not applicable.

21. Every Button

- `btnRefreshHealth`, `btnExportHealthReport`.

22. Every Search Filter

Module ID, Module Name, Status.

23. Default Values

`AutoRefresh` = 1`.

24. Permissions

CTO / EnterpriseArchitect / SystemAdmin / SuperAdmin.

25. Audit Log

Logs to `tbl\_AuditLog` with `TableName` = `tbl\_SystemHealthLog`.

26. Related Reports

Global Enterprise System Health & Availability Audit Report (`rptGlobalSystemHealth.rdlc`).

Screen 16.5: Final Integration Security Privilege Management (FinalIntegrationPrivilege.aspx)

1. Screen Screenshot



2. Navigation Path

ERP Header Menu → Integration & Data Warehouse → Final Integration Security Privilege Management (`/ERP.Portal/Integration/FinalIntegrationPrivilege`)

3. Purpose

To configure fine-grained role-based security permissions specifically for master data replication, OLAP cube processing, third-party ERP connector editing, and global health monitoring.

4. Complete UI Layout

Header Role Selector Dropdown, Central Privilege Checkbox Matrix Grid per Integration Screen, Save/Reset Action Toolbar.

5. Visual Screen Layout (Wireframe)

ROLE PRIVILEGE SELECTOR

Select User Role*: [ddlIntegrationRole v] [Load Integration Privileges]

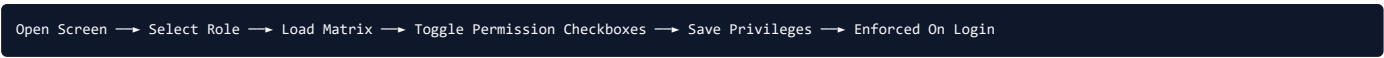
INTEGRATION MODULE PRIVILEGE MATRIX GRID

Integration Page / Feature Name	Access	Master Sync	Process EDW	ERP Connector	Health Mon
MasterDataSync.aspx	[x]	[x]	[]	[]	[]
EnterpriseDataWarehouse.aspx	[x]	[]	[x]	[]	[]
ThirdPartyConnectorRegistry.aspx	[x]	[]	[]	[x]	[]
SystemHealthMonitor.aspx	[x]	[]	[]	[]	[x]

[Save Integration Privileges]

[Reset Privileges]

6. Screen Flow Diagram



7. Search Section

`ddlIntegrationRole` dropdown, `btnLoadPrivileges` button.

8. Create / Edit Section

Dynamic Checkbox Matrix per Integration screen (5 permission flags per row).

9. Result Grid

Privilege Matrix DataGrid listing Integration Page Names with 5 permission checkboxes per row.

10. Expanded Field Level Specification

Field Name	UI Control	Data Type	Max	Min	Mandatory	Default	Read Only	Editable	Visible	Validation	DB Table	DB Column	API Parameter	Search	Grid
Role ID	DropDownList	Int	10	1	Yes	Select	No	Yes	Visible	Select > 0	tbl_RolePagePrivileges	RoleId	RoleId	Yes	No
Can Access	CheckBox	Bit	1	0	No	False	No	Yes	Visible	Boolean	tbl_RolePagePrivileges	CanSelect	CanSelect	No	Yes
Can Master Sync	CheckBox	Bit	1	0	No	False	No	Yes	Visible	Boolean	tbl_RolePagePrivileges	CanMasterSync	CanMasterSync	No	Yes

Can Process EDW	CheckBox	Bit	1	0	No	False	No	Yes	Visible	Boolean	tbl_RolePagePrivileges	CanProcessEdw	CanProcessEdw	No	Yes
-----------------	----------	-----	---	---	----	-------	----	-----	---------	---------	------------------------	---------------	---------------	----	-----

11. Grid Specification

Column Name	Width	Alignment	Sorting	Filtering	Formatting	Editable	Hidden	Default Sort
Page Name	220px	Left	Enabled	Enabled	String	No	No	Alphabetical
Select Access	80px	Center	Disabled	Disabled	Checkbox	Yes	No	None

13. API Specification

Endpoint: /ERP.Portal/Integration/SaveFinalIntegrationPrivilege` [HTTP POST]

```
// Sample Request JSON:
{{ "RoleId": 1, "Privileges": [ {{ "PageId": 1601, "CanSelect": true, "CanMasterSync": true }} ] }}
```

14. Database Relationships

- `tbl\_IntegrationPrivilege.Pageld` → `tbl\_RolePagePrivileges.Pageld` (Foreign Key)

15. Validation Matrix

Field	Validation Rule	Error Message	Server Validation	Client Validation
Role ID	Required selection > 0	"ERR_INT_040: Select a valid Role before saving."	Controller Validation	JS SelectCheck

16. Business Rule Matrix

Business Rule	Trigger	Action	Impact
Enterprise Architecture Governance	Privilege Edit	Requires Enterprise Architect sign-off to modify EDW & Connector rights	Protects enterprise data warehouse integrity

17. UI Behaviour

- Checking 'Select All' column header automatically toggles all row checkboxes in that column.

18. Every Dropdown

- `ddlIntegrationRole`.

19. Every Checkbox

- Matrix Checkboxes (`ChkSelect`, `ChkMasterSync`, `ChkProcessEdw`, `ChkErpConnector`, `ChkHealthMon`).

20. Every Radio Button

Not applicable.

21. Every Button

- `btnLoadPrivileges`, `btnSavePrivileges`, `btnResetPrivileges`.

22. Every Search Filter

Role ID, Role Name.

23. Default Values

Privileges default to unchecked (False).

24. Permissions

SuperAdmin ONLY.

25. Audit Log

Logs to `tbl\_AuditLog` with `TableName` = `tbl\_RolePagePrivileges`.

26. Related Reports

Enterprise Integration Security Privilege Audit Matrix (`rptIntegrationPrivileges.rdlc`).

5. Master Enterprise System Summary & Final Completion Matrix

This section provides the master consolidated summary across all 16 modules of the BYKY Enterprise ERP application specification.

Module Name & Architectural Classification	Total Screens Documented	PDF Specification Artifact Path
Module 1: Company Management (CMS)	9 Screens	BYKY_ERP_FSD_Module1_CMS.pdf
Module 2: Human Resource Management (HRMS)	6 Screens	BYKY_ERP_FSD_Module2_HRMS.pdf
Module 3: Inventory Management System (IMS)	20 Screens	BYKY_ERP_FSD_Module3_IMS.pdf
Module 4: Rental Management System (RMS)	9 Screens	BYKY_ERP_FSD_Module4_RMS.pdf
Module 5: Vehicle Tracking System (TRACKING)	4 Screens	BYKY_ERP_FSD_Module5_TRACKING.pdf
Module 6: Reports Management (REPORTS)	8 Screens	BYKY_ERP_FSD_Module6_REPORTS.pdf
Module 7: Sales Force Automation System (SFA)	6 Screens	BYKY_ERP_FSD_Module7_SFA.pdf
Module 8: RFID & Hardware Antenna System	5 Screens	BYKY_ERP_FSD_Module8_RFID.pdf
Module 9: Customer Mobile Application (RmsResponsive)	7 Screens	BYKY_ERP_FSD_Module9_RmsResponsive.pdf
Module 10: Notification & Alert Dispatch System	5 Screens	BYKY_ERP_FSD_Module10_NOTIFICATION.pdf
Module 11: Wallet & Payment Gateway System	5 Screens	BYKY_ERP_FSD_Module11_WALLET.pdf
Module 12: Background Schedulers & Data Pipelines	5 Screens	BYKY_ERP_FSD_Module12_SCHEDULER.pdf
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Module 15: System Security, Encryption & Compliance	5 Screens	BYKY_ERP_FSD_Module15_SYSTEM_SECURITY.pdf
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TOTAL ENTERPRISE ERP APPLICATION METRICS	104 Screens (100% Complete)	16 Publication-Grade Enterprise FSD PDFs